

Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** Python, Go, Java, TypeScript, JavaScript, SQL
- **AI & Data:** RAG Architecture, Vector Databases (FAISS/ChromaDB), LangChain, OpenAI API, Anthropic API
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC
- **Infrastructure & DB:** AWS (EC2), Docker, PostgreSQL, MongoDB, Linux

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 – Present

Personal Project · Python, Go, LangChain, FAISS

- Automated end-to-end job application workflows by architecting an autonomous AI Co-Pilot in Python, enabling the discovery and parsing of thousands of postings via concurrent JobSpy scrapers.
- Eliminated hallucinations in resume generation by building a RAG engine using LangChain and OpenAI APIs, delivering **production-quality code** for ATS-optimized PDFs based on verified data.
- Guaranteed application state durability and fault tolerance across network failures by developing TitanStore in Go, implementing the Raft consensus protocol and a custom WAL, resulting in a resilient distributed memory bank.
- Enabled seamless scaling to 100+ users by designing a high-throughput TCP client API, allowing Python AI workers to perform concurrent, real-time status updates to the Go database cluster.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 – Jan 2026

Personal Project · Python, Docker, AWS, FastAPI

- Optimized deployments and reduced infrastructure costs by implementing **full-stack application development** practices to migrate a platform from Azure to **AWS services** (EC2) using **Docker / Kubernetes** container orchestration, resulting in improved system efficiency.
- Converted unstructured prerequisite text into a deterministic Abstract Syntax Tree (AST) by designing and implementing a custom parser to tokenize nested boolean strings, resulting in precise prerequisite evaluation.
- Increased data throughput by 10x by engineering a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, resulting in faster data acquisition while respecting server constraints.
- Optimized read performance for complex queries by designing a normalized schema using SQLAlchemy with JSON-type columns to store recursive tree structures, resulting in reduced query latency.

Gridlock Casino (2D Arcade Engine) Feb 2026 – Present

Collaborative Project · Java, Swing, Maven, JUnit

- Maintained a consistent 60-FPS rendering cycle and concurrent user input by architecting a custom 2D grid-based arcade engine in Java using a strict MVC architecture, resulting in a smooth user experience.
- Ensured core engine reliability and logic accuracy by implementing comprehensive JUnit testing and managing dependency resolution via Apache Maven, resulting in a stable build for a 6-developer team.

n54 Present

Personal Project · config, github-config

- Standardized GitHub profile presentation by managing configuration files within the n54 repository, resulting in a professional public presence.
- Improved profile maintainability by organizing github-config settings, resulting in consistent display across the platform.

EDUCATION

Bachelor of Science, Computing Science

May 2025 – Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).

- Langara College | Associate of Science, Computer Science Jan 2024 – Apr 2025